DATA 620 Week 4: Centrality Measures

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1 DATA 620 – Week 4 Assignment: Centrality Measures

- **Author:** Kerwin Charles
- **Dataset:** SNAP *ego-Twitter* (Stanford Large Network Dataset Collection)
- **Builds on:** Week 3 Graph Visualization (same ego network, user `374007416`)

1.1 Overview

The idea behind this week's assignment is that centrality measures can be used to predict positive or negative outcomes for a node. The task is to find network data on the web that has at least one categorical variable for every node, explain how I would load it, and describe an outcome that could be predicted by comparing degree centrality across the categories.

In Week 3 I worked with one Twitter ego network from SNAP. Because I already know that network well, I am staying with it. The piece that was missing last week was a categorical variable, and this dataset has one built in: the **circles**, which are the Twitter lists the ego user sorted the people they follow into. Every person in the network is either on at least one of those lists or on none of them, so every node can be labelled **Listed** or **Unlisted**.

This notebook covers:

1. the data source and why I chose it,
2. the nodes, edges and categorical variables,
3. a high-level plan for loading the data, with a short check that the plan works,
4. how degree centrality applies to this network,
5. the hypothetical outcome I expect degree centrality to predict, and
6. the limitations of the data.

1.2 Data source

Where it comes from. The *Social circles: Twitter* dataset is published by the Stanford Network Analysis Project (SNAP). It was collected for a 2012 study on discovering social circles, using Twitter data gathered from public sources. It includes the ego networks, the circles and profile features for each user.

Size. Across all 973 ego networks the combined graph has **81,306 users** and **1,768,149 directed follow edges**.

How it is accessed. Everything is a plain HTTP download, so it can be pulled straight into Python:

File on SNAP	Contents
`twitter.tar.gz`	the 973 ego networks, five files per ego (about 22 MB compressed)
`twitter_combined.txt.gz`	the edges from all ego networks combined
`readme-Ego.txt`	description of the file formats

Why not the live Twitter API? In February 2023 Twitter ended free access to its API and moved to paid tiers. Collecting a fresh follow network by snowball sampling through the API (starting from one user and following their connections outward) is no longer practical for a class project. The SNAP archive is a public snapshot of the same kind of data.

Why this dataset is a good fit

- Every edge means the same thing ("A follows B"). Mixing different kinds of relationships in one network makes centrality hard to interpret.
- It is a one-mode network (users to users). Hashtags are only used as node attributes, not as nodes, so I avoid the two-mode (bipartite) problem, where centrality is not comparable across two node types.
- The circles give a categorical variable that the ego created, rather than one I have to invent. The IDs are anonymised, which makes them harder to interpret, but the hashtags and circles give the nodes meaning that plain IDs don't have.

1.3 Nodes, edges and categorical variables

Each ego network is stored as five text files named after the ego's user ID:

File	Format	Meaning
`<ego>.edges`	`a b` per line	directed edge: user **a follows b** (both followed by the ego)
`<ego>.circles`	`circle_id member member ...`	one line per circle (Twitter list) made by the ego
`<ego>.feat`	`node 0 1 0 ...`	binary features for each user
`<ego>.featnames`	`index name`	feature names: hashtags (`#`) and mentions (`@`)
`<ego>.egofeat`	`0 1 0 ...`	the ego's own features

- **Nodes:** the users the ego follows (the *alters*). The ego is not written in the edge file and is assumed to follow every node. As in Week 3, I leave the ego out, because a node connected to everyone turns the network into a hub-and-spoke shape (in Week 3 it shrank the diameter from 8 to 2).
- **Edges:** directed "follows" relationships among the alters. They are unweighted.

Categorical variables

Variable	Values	Built from	Every node has a value?
List status (primary)	Listed / Unlisted	in at least one circle vs. in none	Yes
Circle	circle ID (can be several)	`<ego>.circles`	Yes ("none" for Unlisted)
Photography tagger (secondary)	Yes / No	has `#photography`, `#photos`, `#photoshop` or `#photographers`	Yes, with a caveat (see the loading checks)

List status is the main variable because it is complete and not ambiguous. A user can be in several circles, so "Circle" on its own is not a single category per node; collapsing it to Listed/Unlisted fixes that.

1.4 Plan for loading the data

High-level steps

1. **Download** `twitter.tar.gz` from SNAP with `urllib` (skipped if the file is already on disk), and extract it into `data/twitter/`.
2. **Read the edges** of the chosen ego network into a directed `networkx` graph (`DiGraph`), one edge per "a follows b" line.
3. **Add every alter as a node**, including people listed in `.feat` who have no edges to the other alters, so nobody is silently dropped (13 such users in this network).
4. **Read the circles** and set a node attribute `list_status` = Listed or Unlisted, plus a `circles` attribute listing the circle IDs.
5. **Read the features**, normalise the hashtags (lower case, trailing punctuation removed) and set the secondary attribute `photo_tagger`.
6. **Validate**: every node has a `list_status`, every circle member is a node, and the group sizes are large enough to compare.
7. **Compare with the wider dataset**: count how many of the 973 ego networks use lists, to see how typical this ego is.

The code below carries out steps 1–7. It only checks that the data can be loaded and labelled; it does **not** compare centrality between the groups.

```
Archive size     : 22.3 MB
Files extracted  : 4865
Ego networks     : 973
File types       : [('.circles', 973), ('.edges', 973), ('.egofeat', 973),
('.feat', 973), ('.featnames', 973)]
```

```
Ego 374007416: 139 users, 638 directed follow edges, 11 circles
```

	count
users (nodes)	139
users with a list_status	139
circle members not found as nodes	0
Listed (in 1+ circles)	109

Unlisted (in no circle)	30
Listed users in 2+ circles	17
photography taggers	24
users with no features at all	20
isolated users (no follows among alters)	13

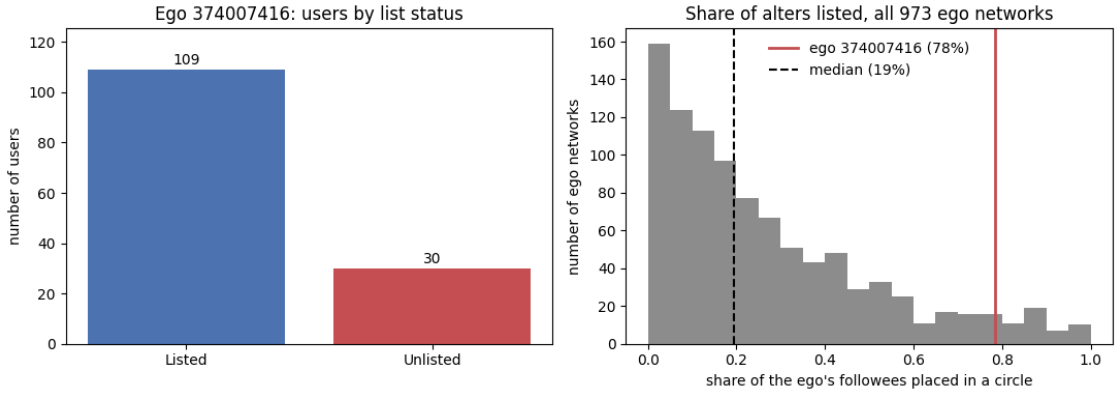
photo_tagger	No	Yes	Total
list_status			
Listed	89	20	109
Unlisted	26	4	30
Total	115	24	139

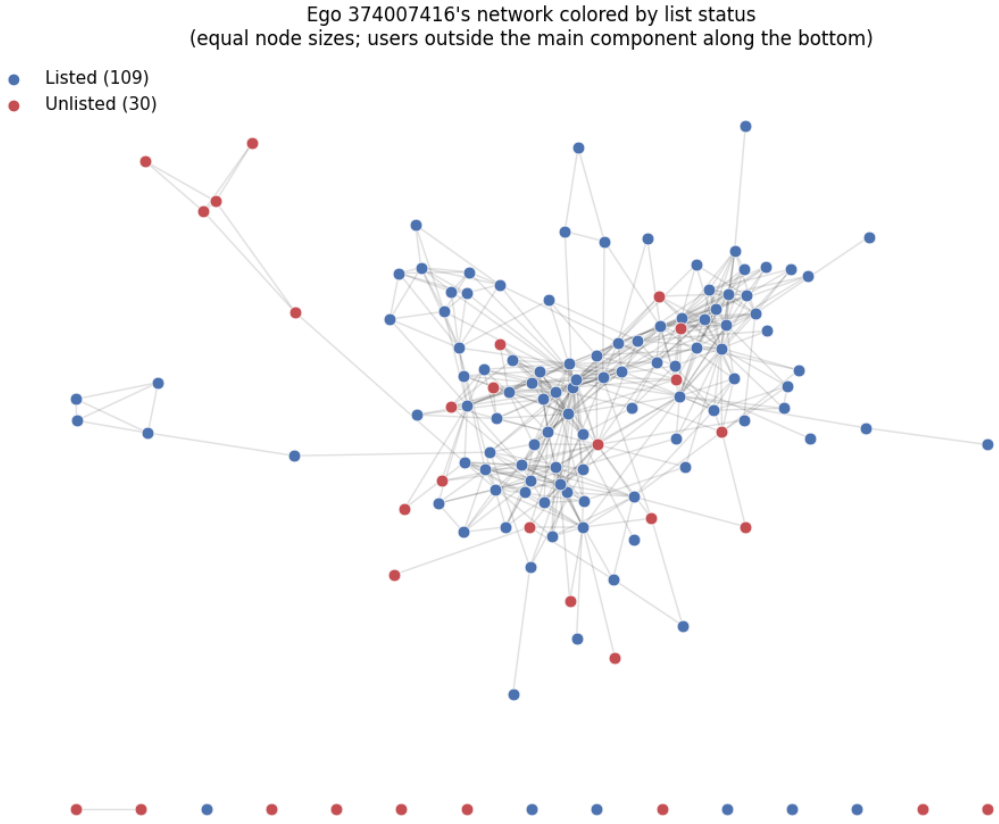
isolated	connected	isolated	Total
list_status			
Listed	103	6	109
Unlisted	23	7	30
Total	126	13	139

What the checks show

- Every one of the 139 users has a list status: **109 Listed** and **30 Unlisted**. Both groups are big enough for a two-sample comparison, although the Unlisted group is small.
- Every circle member is a node in the network (0 missing), so nobody on a list is lost.
- 17 of the Listed users are in two or more circles, which confirms that circle ID on its own can't be a single category per node.
- The two variables overlap only partly: 20 of the 24 photography taggers are Listed.
- The photography variable needs care: 20 users have **no features at all**, so a "No" can mean "doesn't use those tags" or "we have no tag data for this user". That is why it is the secondary variable and not the main one.
- 13 users have no follow links to anyone else in the group (6 Listed, 7 Unlisted). Their degree is 0. I will keep them in the degree comparison because that is real information about them.

	value
ego networks	973
with at least one listed alter	938
with both Listed and Unlisted alters	936
median alters per ego network	140
median share of alters listed	19%
share listed in ego 374007416	78%





What the wider check shows

- The download, extraction and labelling all work: the archive holds 973 ego networks (4,865 files, five per ego).
- Lists are common: 938 of the 973 egos put at least one of the people they follow on a list.
- Ego `374007416` is **not typical**. The ego listed 78 % of the people they follow, while the

median ego network listed only 19 %. That is worth keeping in mind before generalising from this one network.

1.5 Degree centrality in this network

What makes a node "important" depends on what a tie means. Here a tie means "A follows B", so a follow is a choice to receive someone's tweets. Because the edges are directed, degree centrality splits into two measures:

Measure	What it captures	Meaning in this network	Role
In-degree centrality	number of incoming ties; finds the "celebrities"	how many other users in the group follow you (popularity)	**main measure**
Out-degree centrality	number of outgoing ties	how many in the group you follow (how active you are)	secondary

- `networkx.in_degree_centrality` and `out_degree_centrality` divide each count by $n - 1$ (138 here), so a score of 0.10 means about 10 % of the other users.
- The 13 isolated users have a degree of 0. That is real information about them, so they stay in the comparison.
- Centrality measures say who is central, not why. The comparison only asks whether the two groups differ.

1.6 Questions and the hypothetical outcome

Main question. Do the users the ego put on a list have a higher in-degree centrality (more followers inside the group) than the users the ego left off every list?

Hypothetical outcome to predict: being added to one of the ego's lists. Being listed is a positive outcome for a user: it means the ego has chosen to curate that account into a group they read. Staying unlisted is the negative outcome. If Listed users have clearly higher in-degree centrality than Unlisted users, then in-degree could be used to predict whether the ego will put a newly followed account on a list. This is close to what the dataset was originally built for: discovering circles automatically from network structure and profile features.

Why I expect Listed users to be more central

- *Homophily.* People tend to connect with people like themselves ("birds of a feather flock together"). If the ego builds lists around shared interests, the people on a list should also follow each other, which raises their in-degree.
- *Week 3 evidence.* Last week every circle with five or more members was 3 to 12 times denser inside than the network as a whole, so circle members are tightly connected to each other.
- *Embeddedness.* Twitter activity during the 2011 Egyptian revolution showed that being embedded in a dense cluster of like-minded people predicted influence (reactions per tweet) better than a large follower count.

Why the answer could be "no"

- In Week 3 the most followed user in the group (`20536157`, 20 followers in the group) follows nobody back and is **not on any list**. If the ego lists friends and leaves "broadcaster" accounts unlisted, the most popular users could sit in the Unlisted group. Comparing out-degree as well as in-degree will show whether listing tracks mutual, active relationships rather than popularity.

How the comparison would work. Compute in-degree centrality for every user, then compare the average for Listed users with the average for Unlisted users. A clearly higher average for Listed users would support using in-degree to predict who gets listed; similar averages would mean in-degree is not a useful predictor here.

Hypotheses

	Statement
H1	Mean in-degree centrality is higher for Listed users than for Unlisted users.
H2	Listed users also have higher out-degree centrality (they follow more of the group), which would point to mutual, active relationships rather than popularity alone.
H3 (secondary)	Photography taggers have higher in-degree centrality than other users in this photography-centred network, i.e. the community's main topic sits at its core.

The null hypothesis for each is that there is no difference between the groups.

1.7 Limitations

- **Old data.** The crawl was published in 2012, so it reflects Twitter at that time.
- **Anonymised IDs.** Users are numbers, so interpretation relies on hashtags and circles.
- **Lists reflect one person's choices.** Circles are the ego's own lists, and egos use them very differently (78 % listed here vs. a median of 19 %).
- **Missing features.** 20 users in this network have no features, which weakens the photography variable.
- **Association, not cause.** Even if Listed users are more central, being central may not be *why* they were listed; both could come from shared interests with the ego.
- **The ego is excluded.** This is deliberate (Week 3), but it means degree only counts follows among the alters, not each user's full Twitter audience.

1.8 References

- McAuley, J. and Leskovec, J. (2012). *Learning to Discover Social Circles in Ego Networks.* Advances in Neural Information Processing Systems (NIPS).
- Stanford Network Analysis Project (SNAP). *Social circles: Twitter.* https://snap.stanford.edu/data/ego-Twitter.html
- Tsvetovat, M. and Kouznetsov, A. (2011). *Social Network Analysis for Startups.* O'Reilly Media.